

Spain

Catalan Competition Authority

The Autoritat Catalana de la Competència (ACCO, Catalan Competition Authority) is an independent and autonomous administrative body of the Government of Catalonia (Spain) whose mission is to safeguard and promote competition in Catalonia.¹⁹

Why ERICCA

For some time now, the ACCO has noted a significant increase in the number of cases, whether complaints or queries, related to public tenders. Collusion between two or more market operators was one of the main competition issues identified in these tenders. Briefly, economic operators decide to collude when presenting their bids to avoid competition between them to ensure they get a better contract than the one they would have got if they had genuinely competed with each other. By using this strategy, the winning bidder gets a windfall profit which it can then use to compensate the other companies that have fraudulently submitted bids in the public tender. This behavior is known as bid-rigging.

Bid-rigging can take very diverse forms in public procurement, but there are five basic types that may occur individually or in combination:

- **Cover bidding** (also called complementary, courtesy, or symbolic): for example, competitors agree to submit bids that are higher than the bid of the previously chosen winner.
- **Bid suppression**: one or more companies refrain from submitting a bid in the tender process or withdraw their bid so that the previously chosen winner's bid will be accepted (alternatively, firms might choose to submit a joint bid even though they could have bid on their own).
- **Market allocation**: competitors decide to share out the market by geographic areas or contracting authorities.
- **Bid rotation**: companies set up rotation schemes for a number of procedures so that they agree on when they will be the winning bidder.

In response to this, the ACCO has developed a new computer tool called ERICCA (Smart Administrative Procurement Collusion Research Tool).

What ERICCA is and what it does

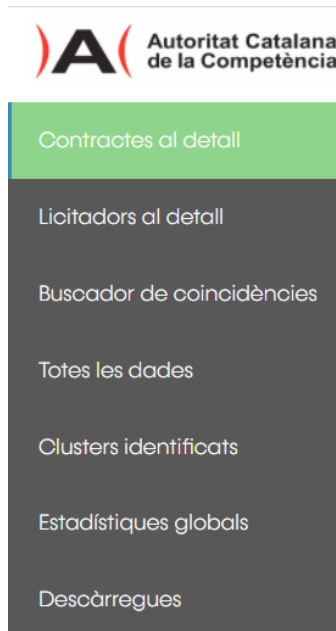
ERICCA is a computer tool that has been developed by the ACCO with two goals:

1. Simplifying the analysis of cases and inquiries in public procurement;
2. Identifying public tenders where participating operators are more likely to have entered into collusive arrangements to submit their bids.

¹⁹ Autoritat Catalana de la Competència, <https://perma.cc/9AzY-U4DY>.

How ERICCA does it

The tool features a menu where the user can choose from a number of 7 options:



The first options are designed to ensure better examination of the cases or enquires sent to the ACCO. Selecting the “Contractes al detall” (Detailed contracts) option provides information about a specific contract (participating companies, winning company, tender budget, bid amount, contracting authority, etc.).

The “Licitadors al detall” (Detailed bidders) option shows all those contracts in which a selected company has participated. It also shows the name of all the companies that it has competed within any of these tenders.

The “Buscador de coincidències” (Match finder) shows all the contracts in which two economic operators who have been selected have taken part.

The “Totes les dades” (All data) option includes all available information.

The “Cluster identificats” (Clusters identified) option is used for proactively identifying potential collusive behaviors. In this case, an Artificial Intelligence algorithm is used. It is an unsupervised machine learning model based on non-Euclidean distances that helps identify clusters between the companies included in the database. This algorithm recognizes companies that are members of the same group, largely because they have taken part in the same tendering procedures, and displays the interrelationships between them as a graphic. Once these clusters have been identified, the tool applies additional filters to detect those where anticompetitive practices are more likely to have taken place. For each cluster, the first filter calculates the average between the tender budget and the award amount. The same filter also calculates the average between the tender budget and the award amount for each company included in the cluster, as well as the number of bids won by each one.

The idea behind applying this filter is to be able to quickly view clusters with differences between the tender budgets and the lowest award amounts, as minimizing the difference between these two variables is one of the main collusion objectives of economic operators. In other words, in the clusters identified that show smaller differences in these variables, it will be more likely in principle that the companies in the cluster have colluded in the tenders in which they participated.



The “Estadístiques Globals” (Global Statistics) option draws on all the information available in the database to show the average discounts (the difference between the tender budget and the bid amount) depending on the selected procedure (open, restricted, negotiation procedure without advertizing, etc.) and the number of participants in the tenders.

Finally, the “Descàrregues” (Downloads) option allows downloading the selected information in CSV format.

Adjudicatari	Descompte mitjà	N. de contractes adjudicats	Descompte mitjà del grup
AL AIR LIQUIDE ESPAÑA, SA	CONFIDENTIAL	3	CONFIDENTIAL
GENERAL ELECTRIC HEALTHCARE ESPAÑA, SA	CONFIDENTIAL	1	CONFIDENTIAL
LINDE GAS ESPAÑA, SA Unipersonal	CONFIDENTIAL	3	CONFIDENTIAL
PALEX MEDICAL, SA, SOCIETAT UNIPERSONAL	CONFIDENTIAL	3	CONFIDENTIAL
PHILIPS IBERICA, SA Societat Unipersonal	CONFIDENTIAL	2	CONFIDENTIAL
SOCIEDAD ESPAÑOLA DE CARBUROS METALICOS, SA	CONFIDENTIAL	4	CONFIDENTIAL

Previous problems

In the previous version, ERICCA had to address three major challenges:

- I. Web scraping has to be used to obtain baseline data and build up a reliable database for ERICCA. However, due to errors generated by this procedure, the data collected were not accurate, and on some occasions the data readout was wrong.
- II. There was no unique identifier for each company since, on the Public Procurement Platform, the name of the same company can be written in several ways. For instance: “Telefónica SL”, “Telefónica Móviles SA.” and “Telefonica Moviles, S.A.”.
- III. Information on the financial bids of all the companies was not directly available on the websites but rather only the bid of the winning company.

Next steps

At the present time, with the aim of facilitating the identification of anti-competitive behaviors between the companies, we are creating new filters for the clusters, which include, for example, for each cluster (1) identification of the winning bidders by contracting authority, and (2) identification of the timeframe of the winning bids.

In the future, we will introduce the information corresponding to the cases instructed by the ACCO so that the tool can learn from them and be more precise (this is reinforced learning).